

1NF

Before normalizing our ERD we initially stored **regulations** inside the **City** table.

city_id	city_name	regulations
1	Barcelona	Short-term rental license, Tourist-rental registration
2	Madrid	Short-term rental license
3	Valencia	Rental registration, Tourist-rental registration

The **regulations** column contained multiple values in one cell. Therefore we decided to split up this table into 3 tables: **City**, **Regulation**, **CityRegulation**. Now the tables are satisfying 1NF, because every cell contains exactly one value.

City

city_id (PK)	city_name
1	Barcelona
2	Madrid
3	Valencia

Regulation

regulation_id	regulation_name
1	Short-term rental license
2	Rental registration
3	Tourist-rental registration

CityRegulation

city_id (PK, FK)	regulation_id (Pk, FK)
1	1
1	3
2	1
3	2
3	3

2NF

Initially we combined city statistics and listings into one table.

city_id (PK)	year (PK)	listing_id (PK)	city_name	avg_rent	social_housing_units	homeless_count	rental_type	listing_price
1	2025	1	Barcelona	1450	20000	3200	Short-term	180
2	2025	2	Barcelona	1450	20000	3200	Long-term	1400
3	2025	3	Madrid	1600	25000	4100	Long-term	1550

We were violating 2NF because the primary key contained 3 attributes, while several columns depended on only part of the key, so we had partial dependencies. (Example: avg_rent depends on city_id + year; rental_type depends on listing_id)

The original table was divided according to what each attribute depends on. This way we ended up having these 3 tables: **City**, **CityStatistics**, **RentalListing**.

City

city_id (PK)	city_name
1	Barcelona
2	Madrid

CityStatistics

city_id (PK, FK)	year(PK)	avg_rent	social_housing_units	homeless_count
1	2025	1450	20000	3200
2	2025	1600	25000	4100

RentalListing

listing_id (PK)	city_id (FK)	platform_id	platform_name	rental_type	listing_amount
1	1	1	Airbnb	Short-term	180
2	1	2	Idealista	Long-term	1400
3	2	2	Idealista	Long-term	1550

3NF

listing_id (PK)	city_id (FK)	platform_id	platform_name	rental_type	listing_amount
1	1	1	Airbnb	Short-term	180
2	1	2	Idealista	Long-term	1400
3	2	2	Idealista	Long-term	1550

Looking at the table **RentalListing** created after 2NF, we have a problem. platform_name doesn't directly depend on the primary key listing_id, instead it depends on platform_id. This is considered a transitive dependency. To solve this problem a separate table **Platform** was formed.

RentalListing

listing_id (PK)	city_id (FK)	platform_id (FK)	rental_type	listing_amount
1	1	1	Short-term	180
2	1	2	Long-term	1400
3	2	2	Long-term	1550

Platform

platform_id (PK)	platform_name
1	Airbnb
2	Idealista